

School mapping in ESRI imagery

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December 12th, 2024



Recap from last meeting

Sampling method choice

- Inverse or Gaussian sampling
- Static or dynamic sampling
- Dynamic or OHEM sampling

- We've seen that static Gaussian sampling and dynamic Gaussian sampling yield the best results

- Our choice: **static Gaussian sampling**

Number of epochs validation

- Increased number of epochs (20 vs 13)
- Measure **F1-score** over epochs on both train and validation set
- Chosen number of epochs for fine-tuning: **20 epochs**

Model fine-tuning

- All Anditi schools + equal number of OSM Vietnam non-schools
- Static Gaussian non-school sampling
- 20 epochs
- Train results
 - **F1: 99.55%**
 - P: 99.80%
 - R: 99.30%

New contributions

Dense inference

- Use fine-tuned model on **overlapping satellite images** (tiles) from chosen districts
- Perform non-maximum suppression (**NMS**) to reduce the number of duplicate detections

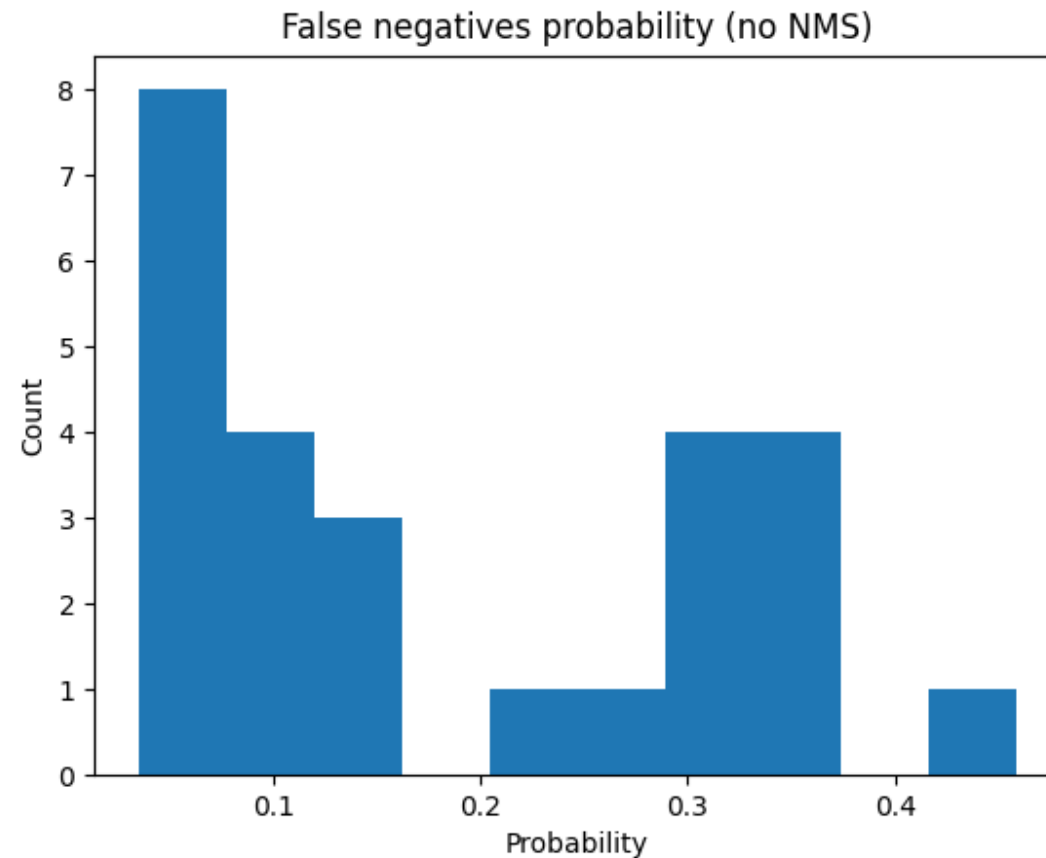
Dense inference

- Measure number of detected Anditi schools:
 - For each Anditi school, do the following:
 - 1. find satellite images which contain the school
 - 2. check if the model's prediction for at least one of the neighboring satellite images is above 0.5
 - 3. if not, memorize the nearest satellite image, as well as the distance and probability
 - Finally, calculate recall, as well as the total number of detected Anditi schools
- Deliver list of potential schools sorted by prediction confidence

Dense inference results (before NMS)

- Found 920 / 946 (**97.25%**) Anditi schools
- Recall (minimum probability): 1.80%
- Recall (mean probability): 47.25%
- Recall (maximum probability): 97.25%
- 26 not detected schools (**2.75%**)

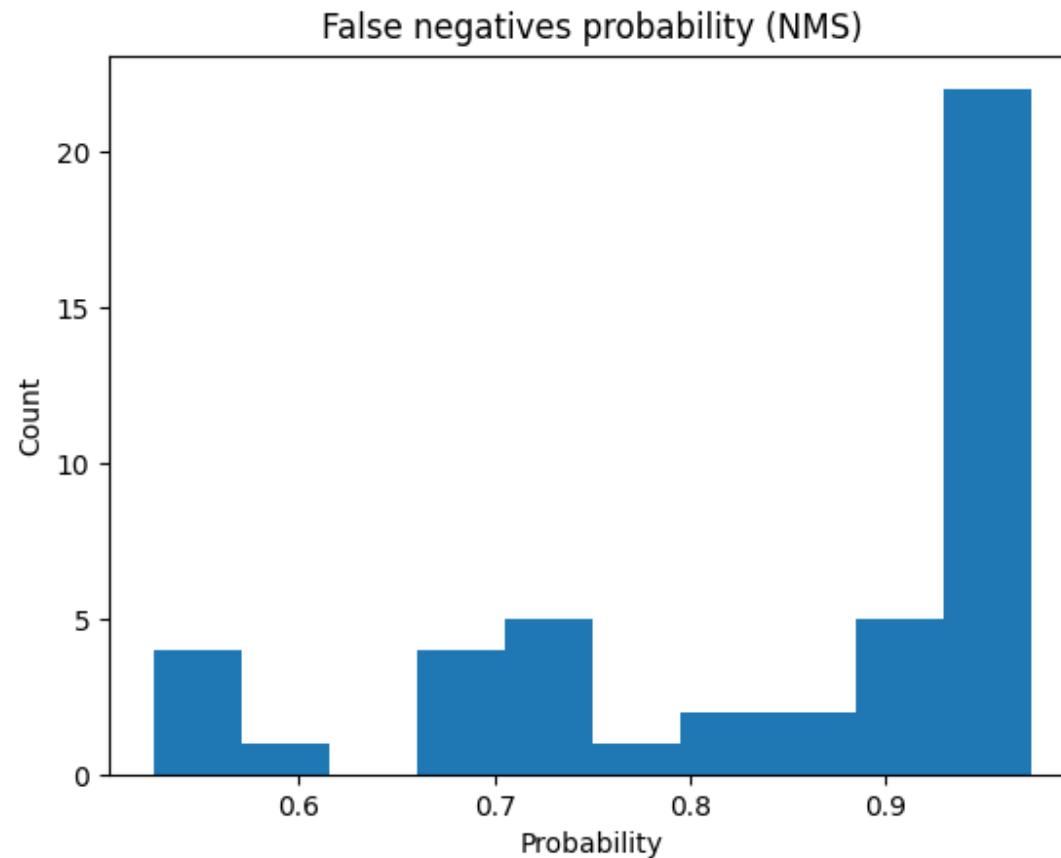
Dense inference results (before NMS)



Dense inference results (after NMS)

- Found 900/946 (**95.14%**) Anditi schools
- Recall (min probs): 95.14%
- Recall (mean probs): 95.14%
- Recall (max probs): 95.14%
- 46 not detected schools (**4.86%**)

Dense inference results (before NMS)



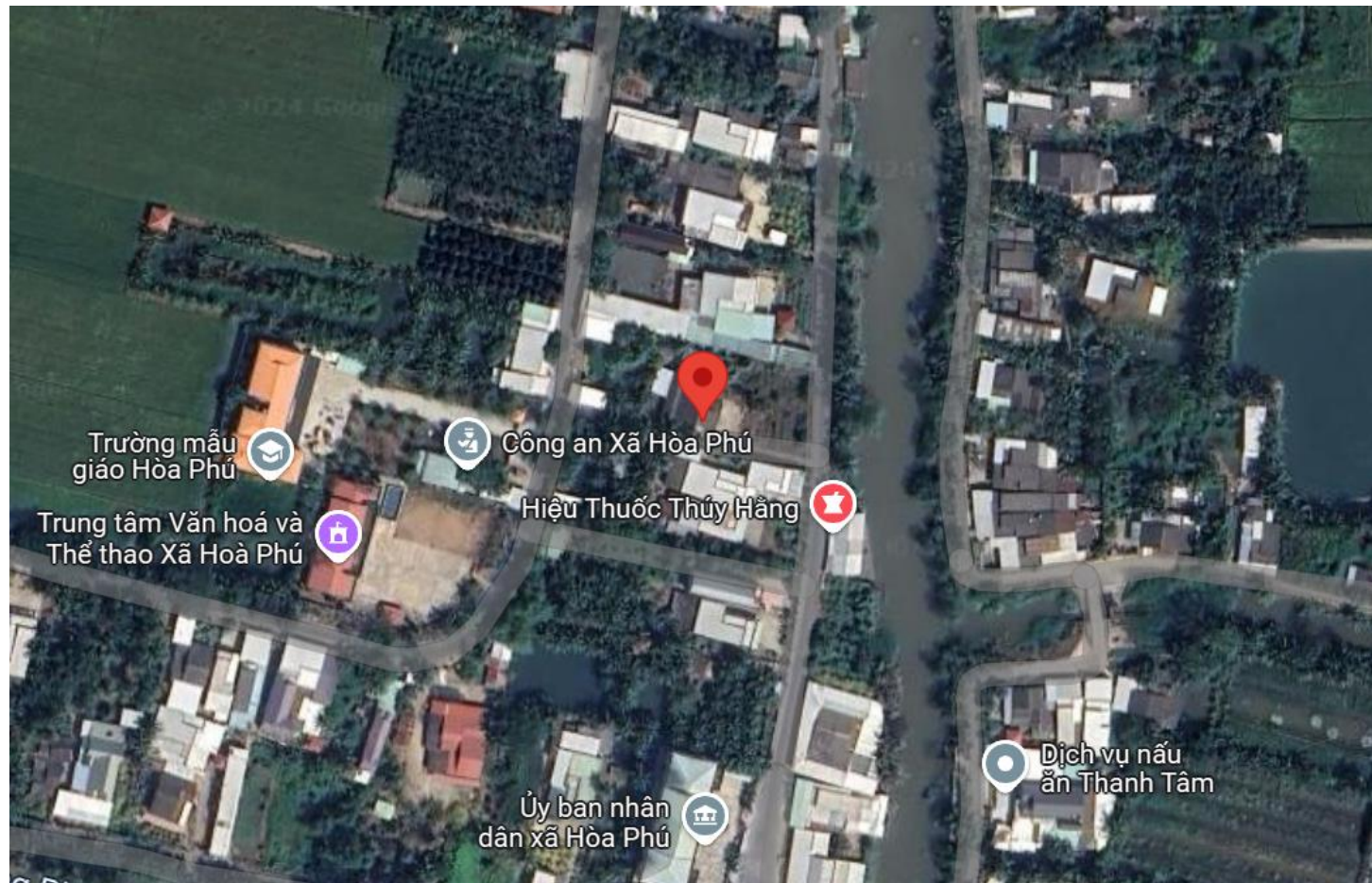
List of potential schools

- Satellite images classified as schools by the fine-tuned model
- Images are sorted by prediction confidence – images with the highest confidence come first
- Non-maximum suppression was applied to the list to avoid overlapping detections

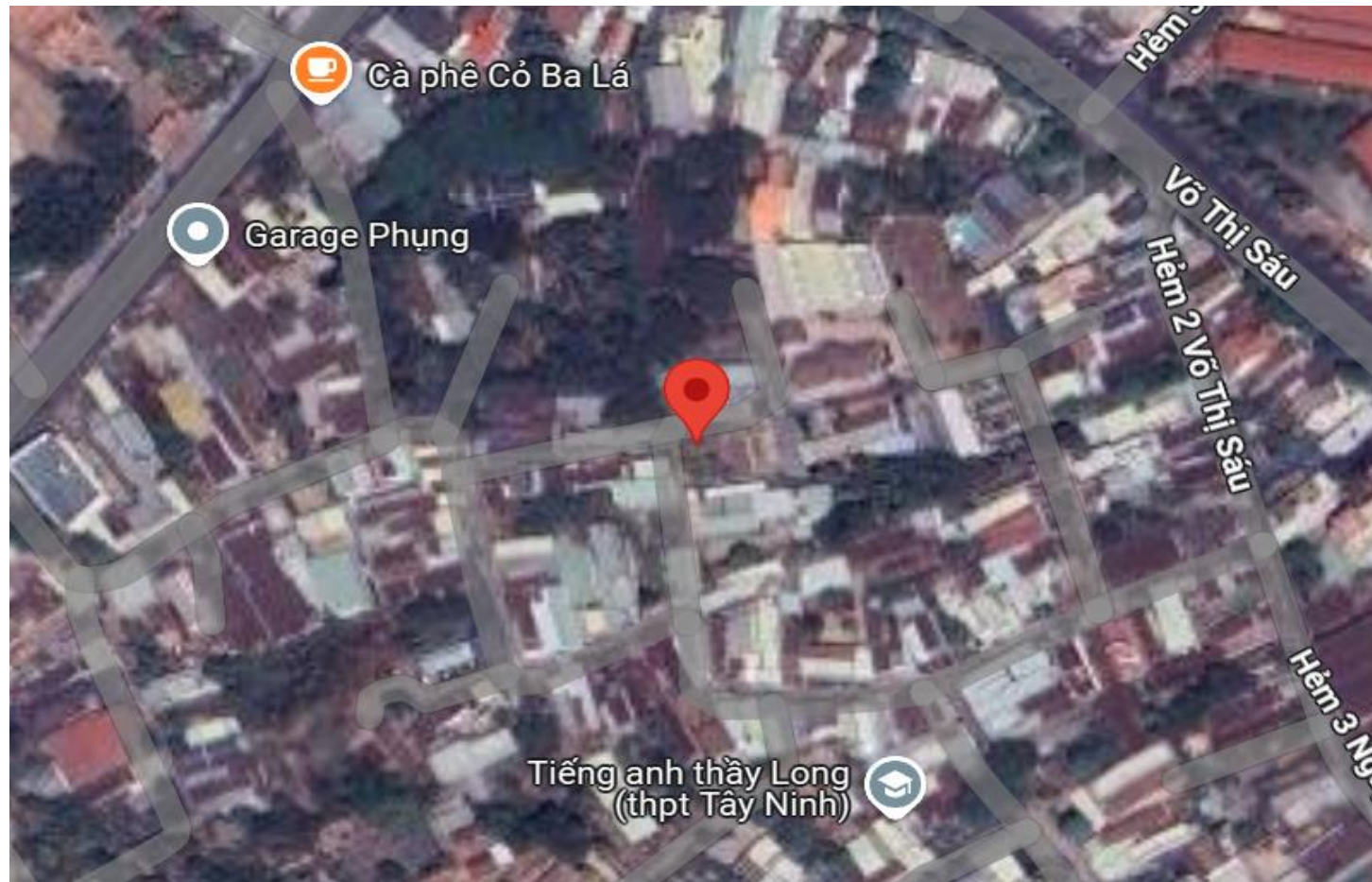
List of potential schools

- 9573 images
- Top 15 images:
 - 4 are marked on Google Maps (manual check)
 - 3 of those 4 images are in the Anditi dataset

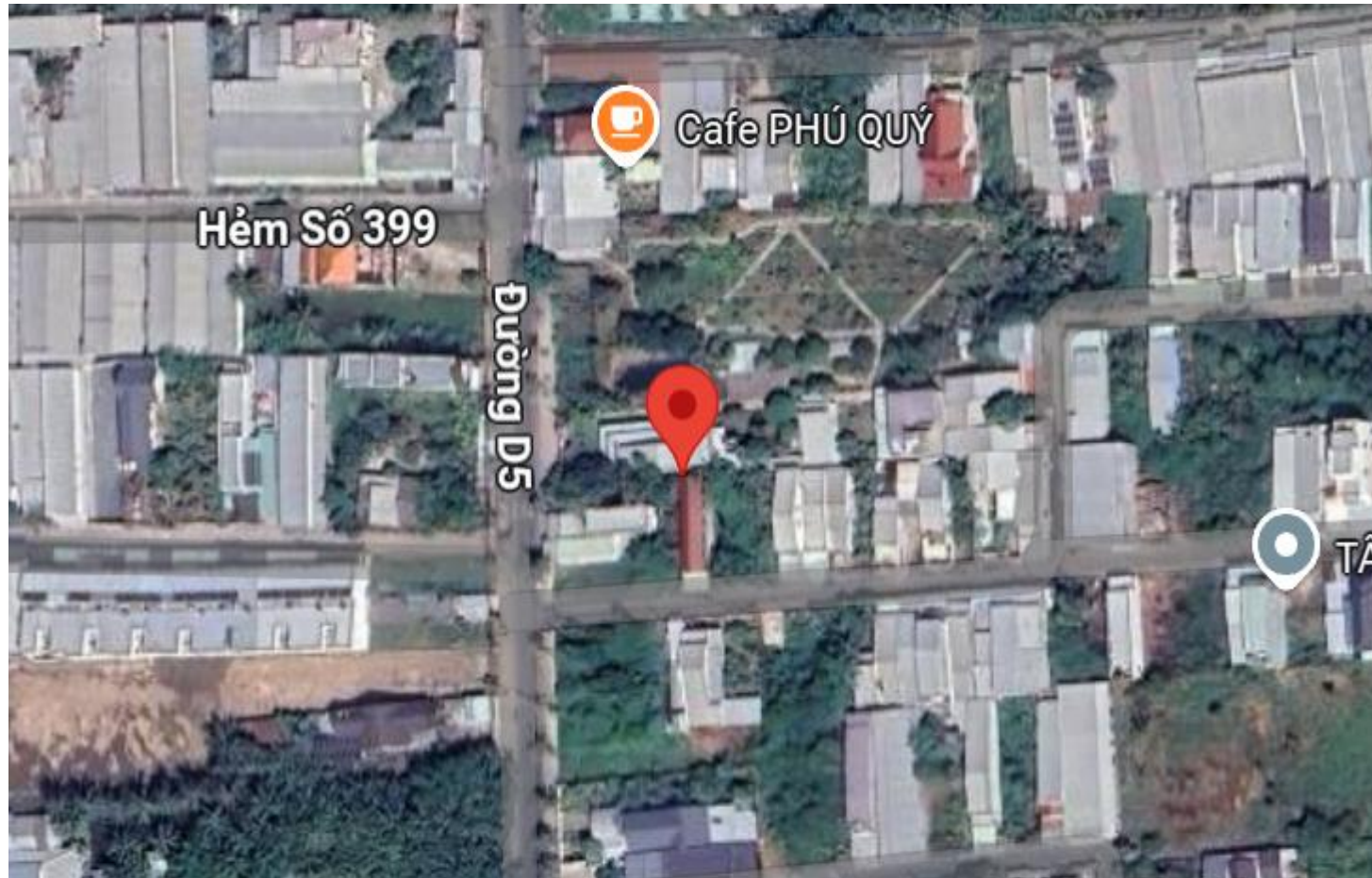
List of potential schools - examples



List of potential schools - examples



List of potential schools - examples



List of potential schools - examples



List of potential schools - examples

